

A Technical Document Simplification System Powered by Artificial Intelligence

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Abstract—In this study, an artificial intelligence system that automatically converts complex technical documents into plain Turkish, understandable by non-expert users, has been developed. The system is designed using the n8n automation platform, the LLaMA 3.1 large language model served by the Groq API, and a PostgreSQL-based Supabase cloud database. Users submit technical text through a web interface and receive simplified output within an average of 2-3 seconds, while all transactions are automatically persisted in the database. The proposed system demonstrates that AI-driven natural language processing applications can be built rapidly and at minimal cost using a no-code/low-code approach.

Keywords—Artificial intelligence, natural language processing, n8n, Supabase, LLaMA, text simplification, automation

I. INTRODUCTION

As technology evolves at an accelerating pace, technical documents have become progressively more complex. Engineering manuals, academic publications, software documentation, and legal texts contain language that is difficult to comprehend for users without domain expertise. This phenomenon contributes to a measurable inequality in access to information and exacerbates the broader digital literacy gap.

Recent advances in large language models (LLMs) have produced significant gains in natural language processing tasks such as text simplification, summarization, and language adaptation. The system presented in this paper combines these capabilities with freely available API services and visual workflow automation tools to deliver a practical end-user application.

The principal aim of this work is to construct an end-to-end AI-powered system without requiring traditional software engineering expertise, following a no-code/low-code methodology. The system is built on top of the n8n automation platform, performs text simplification using Groq's free LLaMA 3.1 model, and persists all records in a Supabase cloud database.

II. SYSTEM ARCHITECTURE

The proposed system follows a modular architecture composed of four interdependent components. Each component fulfills a distinct responsibility and

communicates with adjacent layers via standard HTTP protocols.

A. Web Interface (Frontend)

The user interaction layer is implemented as a single-page application using HTML5, CSS3, and vanilla JavaScript. A modern Aurora-themed visual design has been adopted, and core user-experience features such as .txt file upload, viewing past records, and copy-to-clipboard have been included. The interface forwards user-submitted text asynchronously to the n8n webhook endpoint via the Fetch API.

B. n8n Automation Workflow

The backbone of the system is a four-node workflow hosted on n8n Cloud. The Webhook node accepts POST requests directed at the /sadele endpoint; the technical text contained in the incoming JSON payload is forwarded to the Groq AI node. The simplified text returned by the model is both written to the database via the Supabase node and returned to the client as JSON through the Response node. Data flow between nodes is defined declaratively using n8n's visual expression language.

C. Groq API and the LLaMA 3.1 Model

In the AI processing layer, the freely available llama-3.1-8b-instant model served by Groq has been adopted. The model is configured via a system prompt that mandates conversion of the input into high-school level plain Turkish, removal of technical jargon, and the use of analogies drawn from everyday life. Owing to Groq's low-latency inference infrastructure, the average end-to-end response time has been measured at 2-3 seconds.

D. Supabase Database

For data persistence, Supabase, an open-source Backend-as-a-Service offering built on PostgreSQL, has been used. A single table named documents has been defined, containing the columns original_text, simplified_text, and created_at. Each simplification operation appends a new row to this table. Integration with n8n is established through the Supabase REST API.

III. APPLICATION FLOW

The system implements an end-to-end data processing pipeline that operates from user input to database persistence with full automation. The flow consists of the following steps: (1) the user enters technical text or uploads a .txt file via the web interface; (2) JavaScript serializes the text into a JSON payload and sends a POST request to the n8n webhook; (3) n8n forwards the text to the Groq API and obtains the simplified version; (4) the resulting text is committed to the Supabase table via an INSERT operation; (5) finally, the simplified text is returned to the client as JSON and rendered in the user interface.

This architecture decouples the database layer from business logic and allows additional language models to be integrated in future iterations. Because the entire stack is composed of serverless components, no additional infrastructure effort is required for scalability.

IV. RESULTS AND EVALUATION

During system testing, technical texts from three distinct domains were used: TCP/IP networking protocols, quantum physics, and symmetric-key cryptography. In every test case, the model successfully simplified the input while preserving its original meaning and explained complex concepts using analogies drawn from everyday experience. For instance, the TCP three-way handshake was rendered by the model as a metaphor for initiating a telephone call.

In terms of performance, end-to-end response times ranged between 2.0 and 3.5 seconds, while database write operations completed without any latency or loss. All free-tier components used in the project (Groq, Supabase, n8n Cloud) operated reliably throughout the testing period.

V. CONCLUSION

In this study, a functional artificial intelligence system that converts technical documents into plain Turkish has been successfully designed, implemented, and presented. The system, built through the integration of n8n, Groq, and Supabase, illustrates an effective example of combining no-code automation tools with modern AI services. The resulting architecture has the potential to democratize access to information in both academic and industrial contexts.

Future work will focus on extending the system through user-feedback-driven model fine-tuning, multilingual support, and image-based document input via OCR integration. The full source code of the project is publicly available on GitHub under an open-source license.

VI. REFERENCES

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